

Lesson 1

Part 1: Individual decision-making

1 Decision making in general (Chapter 1)

The task is to understand decision-making (d.m.) at the individual level, i.e. the level of the economic agent, or the decision maker.

But we limit our scope to the economic agent operating in a market economy.

Chapters 2–3, or the consumer theory: given prices and income, what type of consumer decisions do we expect?

2 Consumer choice: the elements of the problem (Chapter 2)

We start by describing some elements of the problem.

1. There is a fixed, finite number L of goods or commodities. Let x_i denote the quantity of good i that is consumed, $i = 1, 2, \dots, L$.

The bundle that the agent consumes is denoted

$$x = (x_1, \dots, x_L).$$

x is a vector, and is supposed to be a column vector.

2. The *consumption set* X : denote the set of possible bundles. We assume quantities are non-negative, i.e. $x_i \geq 0$, $i = 1, 2, \dots, L$. We imagine the quantities are continuous:

$$X = \mathbb{R}_+^L.$$

3. The agent (consumer) lives in a market economy. Let w denote wealth and p_i the price of good i , $i = 1, 2, \dots, L$. Let $p = (p_1, \dots, p_L)$ (the price vector); assume prices are all positive, $p \gg 0$, and

$$p \cdot x = \sum_{i=1}^L p_i x_i \leq w.$$

3 The budget set and its properties

The budget set is the set of affordable or feasible bundles, denoted B :

$$B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}.$$

The subset of bundles where the entire income is spent is

$$\{x \in \mathbb{R}_+^L : p \cdot x = w\},$$

which lies on the *budget hyperplane* $\{x \in \mathbb{R}^L : p \cdot x = w\}$; when $L = 2$ it is the *budget line*.

Properties of the budget set

- Given $w > 0$, B is non-empty, since $x = 0$ is affordable.
- Given $p \gg 0$, B is bounded — a limit to what can be afforded, since $0 \leq x_i \leq w/p_i$ for every good i — and B is a closed set (it is defined by weak inequalities).
- B is a compact set: closed and bounded in $\mathbb{R}^L$.
- B is a convex set: if $x, x' \in B$, then

$$\tau x + (1 - \tau)x' \in B, \quad \tau \in [0, 1],$$

since $\tau x + (1 - \tau)x' \geq 0$ and $p \cdot (\tau x + (1 - \tau)x') = \tau p \cdot x + (1 - \tau)p \cdot x' \leq \tau w + (1 - \tau)w = w$.

4 Two ways to proceed

We have described the economic environment. We can proceed in two ways:

- *The classical approach.* Start from axioms on preferences and see what predictions follow.
- *The revealed-preference approach.* Start from choice-observable data, and infer something about the consumer from these choices.

Transcription notes

- The first page of the original is a blank template page; it is not reproduced here.
- “But we our scope...” had no verb; read as *we limit* our scope.
- “there’re fix # of goods” read as *a fixed number of goods*; the sentence repeated itself (“a fixed number of goods, and a finite number L of goods”) and is merged into one clause.
- **Added:** the one-line reasons behind boundedness, closedness, compactness and convexity of the budget set.
- Index range: the original wrote $i = 1, 2, 3$ for signs and $i = 1, 2, \dots, L$ for prices and goods; with L goods the range $1, \dots, L$ is used throughout.
- “The bundle that the agent *is* consumed” → *that the agent consumes*.
- “none empty” → *non-empty*; “branded” → *bounded*; “afforde” → *afforded*; “closedcet” → *closed set*.
- “The budget line $L = 2$ on the budget hyperplane”: the glyph is clearly a “2”, i.e. the number of goods. The previous transcription treated $\mathcal{L}$ as a name for the set; it now reads (as in MWG) that the set $\{p \cdot x = w\}$ is the budget hyperplane, a budget *line* when $L = 2$.
- The opening line starts with an abbreviation read as “F.S.” (possibly “First”); it is dropped.
- Section headings carry the chapter labels from the original (“chapter 1”, “chapter 2”).
- “ $Tx + (1 - T)x'$ ” read as $\tau x + (1 - \tau)x'$, with $\tau \in [0, 1]$.
- “fiscal envior.” is illegible; read as *the economic environment*.
- “Axioms” → *axioms*.